

PINCHY / physical parts and connections

SPH0645 • DFRobot DFR0954 • Akyga LP503759 / 1350 mAh • bench-test proposal

Waveshare ESP32-S3-Touch-LCD-2.1B

Rear PCB; reference photo for the shared 2.1/2.1B design



Orient by BAT, USB and microSD; verify harness polarity.

J1 • BAT

MX1.25 2-pin battery connector

- 1 BAT+
- 2 GND / BAT-

Pin numbers, not a connector-face view.

Akyga • LP503759 / 1350 mAh



3.7 V • 1350 mAh • PCM • factory JST 2.54 mm

Audio mode: disconnect USB-UART.

I²S uses GPIO19/20. Native USB power requires D+ and D- individually isolated. Firmware must release GPIO43/44 and USB.

* J9.11: docs say NC, older schematic says GND — leave open. The J9 expansion does not show the connector face.

J9 • SH1.0 • 12 pin

Pin numbers from the schematic

- 12 GPIO0
- 11 NC / GND*
- 10 RXD / GPIO44
- 09 TXD / GPIO43
- 08 SDA / GPIO15
- 07 SCL / GPIO7
- 06 3V3
- 05 GND
- 04 D+ / GPIO20
- 03 D- / GPIO19
- 02 VBUS / 5V
- 01 GND

Waveshare VCC

Net at U5 VIN / C14.
Physical solder point still to be identified.
This is not the 3V3 pin.

Replace the plug first

MX1.25 2-pin to J1

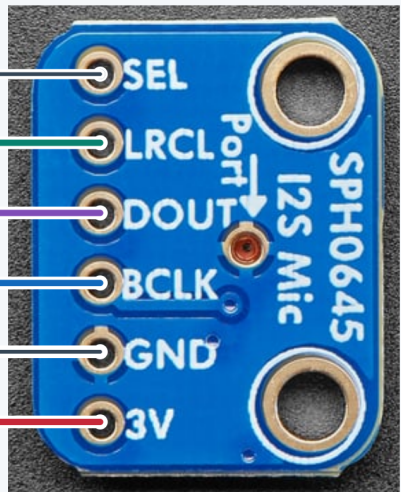
- BAT+ → J1.1
- BAT- → J1.2

Check polarity with a meter.

Factory JST 2.54 mm plug
replace with MX1.25.

Adafruit SPH0645 • #3421

Sound-port side of the microphone



Keep the "PORT" opening clear.

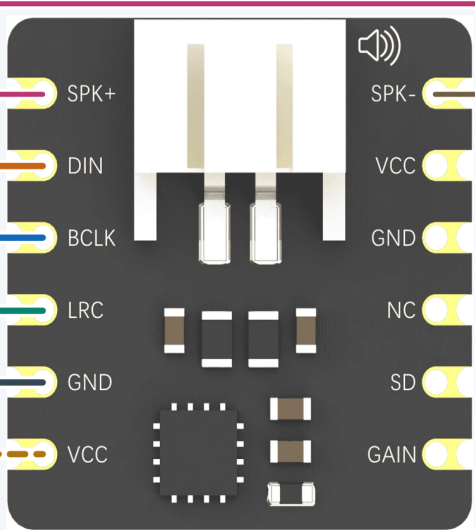
RPD1 • 100 kΩ

DOUT ↔ GND
One lead to each of these nets.
Keep the DOUT wire continuous.
Solder near the microphone pads
and insulate the resistor leads.

SEL ↔ GND: left channel.

DFRobot DFR0954 • MAX98357A

Amplifier variant from the shopping list



Leave GAIN and SD open. Do not use NC.
Duplicate VCC / GND pads: use one pair.

Speaker • 8 Ω / 1 W

Kamami 560816 • factory leads



SPK- is not ground.

Neither speaker lead connects to GND.
Start at low volume (1 W speaker).

Dashed power connections: verify before connecting

Battery: max. charge 1.35 A; standard charge 270 mA, fast charge 675 mA.
Waveshare schematic: R7 = 82 kΩ → approx. 2 A setting. Adjust and measure first.
Amplifier: verify VCC, its 3.3–5 V range and current capacity, including on battery.

• junction / branch Crossing wires without a dot are not connected.